MTE 203 – Advanced Calculus
Week 2 Homework

Curves and Surfaces (S.11.2)

Problem 1 [S. 11.2, 37]:

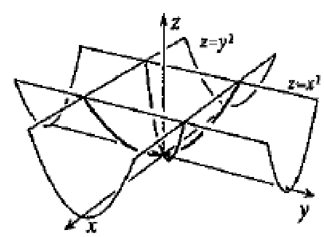
Draw the curve defined by the equations $x + 2y = 6$, $y - 2z = 3$

Problem 2 [S11.2, 45]:

Draw the curve defined by the equations $z = x^2$, $z = y^2$

Solution:

The given equations represent to parabolic cylinders. One opening along the y-axis and one opening along the y axis.



To aid visualization, it is often helpful to first draw the projections of the cylinders onto the coordinate planes. The projection of the cylinder $z = x^2$ can be easily drawn in the $x - z$ plane, as the unconstrained variable for this surface is "y". Similarly, the projection of the cylinder $z = y^2$ can be easily drawn in the $y - z$ plane, as the unconstrained variable for this surface is "x".

Problem 3 [S.11.2, 49] :

Find the equation for the projection in the $x - y$, $y - z$, and $x - z$ planes and draw the curve given by the following equations.

$$x^2 + y^2 = 4, \qquad y = x$$

Problem 4 [S.11.2, 61] :

Find the equation for the projection in the x-z plane and draw the curve given by the following equations.

$$x^2 + y^2 - 2y = 0, \qquad z^2 = x^2 + y^2$$

Parametric and Vector Representations of Curves (S.11.10)

Problem 1 [11.10; Prob. 13]

Draw and plot the curve with the given position vector,

$$\vec{r}(t) = (t-2)\hat{\imath} + (2-3t)\vec{\jmath} + 5t\hat{k}$$

Problem 2 [11.10; Prob. 15]

Draw and plot the curve with the given position vector,

$$\vec{r}(t) = (\cos t)\hat{\imath} + (\sin t)\vec{\jmath} + (\cos t)\hat{k}, \quad 0 \le t \le \pi$$

Extra Practice Problems[1]

The answers to these problems, along with all even-numbered exercises, are provided at the back of the textbook. Detailed solutions can be found in Trim's Student Solution Manual. If you don't own a copy, a reserved edition is available at the Davis Centre.

1. S. 11.2, Probs. 14, 44, 54, 68
2. S. 11.10, Probs. 14

[1] You are strongly encouraged to try and solve the problems assigned in this homework on your own, before looking at the solutions. These problems are only the minimum necessary to master the material. You are also encouraged to do additional problems from the text for practice (some suggested **extra practice problems** have been listed here).